

FIITJEE

ALL INDIA TEST SERIES

PART TEST – II

JEE (Main)-2022

TEST DATE: 11-12-2021

Time Allotted: 3 Hours

Maximum Marks: 300

General Instructions:

- The test consists of total 90 questions.
- Each subject (PCM) has 30 questions.
- This question paper contains **Three Parts**.
- **Part-A** is Physics, **Part-B** is Chemistry and **Part-C** is Mathematics.
- Each part has only two sections: **Section-A** and **Section-B**.
- **Section – A** : Attempt all questions.
- **Section – B** : Do any five questions out of 10 Questions.

Section-A (01 – 20, 31 – 50, 61 – 80) contains 60 multiple choice questions which have **only one correct answer**. Each question carries **+4 marks** for correct answer and **-1 mark** for wrong answer.

Section-B (21 – 30, 51 – 60, 81 – 90) contains 30 Numerical answer type questions with answer XXXXX.XX and each question carries **+4 marks** for correct answer. There is no negative marking.

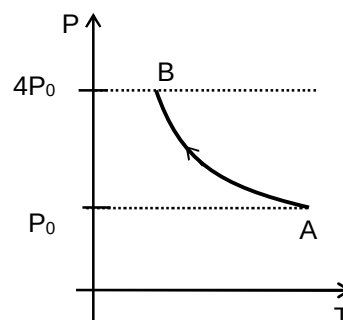
Physics

PART – A

SECTION – A (One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

1. Three moles of an ideal diatomic gas is taken through a process AB as shown in the figure. During the process, pressure and temperature of gas varies such that $PT^2 = \text{constant}$. If $T_B = 300$ K, the work done on the gas in the process is



- (A) $-900 R$
 (B) $+2700 R$
 (C) $-2700 R$
 (D) $+900 R$

Ans. B

Sol. $PT^2 = \text{constant}$
 $\Rightarrow T_A = 2T_B = 600$ K
 Or $P^3V^2 = k$
 $\Rightarrow P = \frac{k^{1/3}}{V^{2/3}}$
 $W_{AB} = \int_A^R PdV = 3k^{1/3} (V_B^{1/3} - V_A^{1/3})$
 $= 3(P_B V_B - P_A V_A) = 3nR(T_B - T_A)$

2. An ideal gas has a molar heat capacity at constant pressure $C_p = \frac{5R}{2}$. The gas is kept in a closed vessel of volume V_0 at a temperature T_0 and pressure P_0 . If $10P_0V_0$ heat is supplied to the gas, then the final pressure of the gas is

- (A) $\frac{23}{3}P_0$
 (B) $\frac{17}{3}P_0$
 (C) $5P_0$
 (D) $3P_0$

Ans. A

Sol. $Q = \Delta U = nC_v \Delta T$

$$\text{Or } 10nRT_0 = n\left(\frac{3}{2}R\right)(T - T_0)$$

$$\Rightarrow T = \frac{23}{3}T_0$$

$$\therefore P \propto T$$

$$\therefore P = \frac{23}{3}P_0$$

3. Three identical particles, each of mass 1 gm and charge 'q' are suspended from a common point by insulated massless strings, each of length 100 cm. If the particles are in equilibrium at the corners of an equilateral triangle of side length 3 cm lying in the horizontal plane, calculate the charge 'q' on each particle. (Take $g = 10 \text{ m/s}^2$)

- (A) 1 nC
(B) 10 nC
(C) 0.1 nC
(D) $\sqrt{10}$ nC

Ans. D

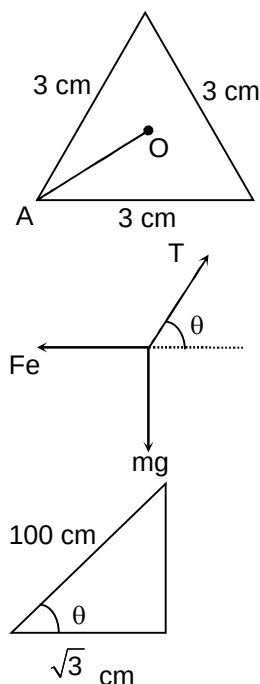
Sol. $AO = \frac{2}{3} \times \frac{\sqrt{3}}{2} \times (3\text{cm}) = \sqrt{3} \text{ cm}$

Net electrostatic force on each particle,

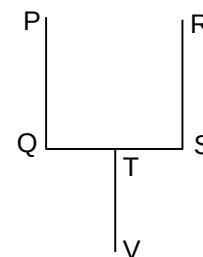
$$F_e = \frac{kq^2}{(0.03)^2} \times 2\cos 30^\circ = \sqrt{3} \times 10^{13} q^2$$

F.B.D.

$$\text{Now, } \tan \theta = \frac{mg}{F_e} \Rightarrow q = \sqrt{10} \text{ nC}$$



4. Four identical rods are joined together as shown in the figure. T is the mid point of the rod QS. Points P, R and V are maintained at 10°C , 15°C and 30°C respectively. Assuming no heat loss to the atmosphere, find the temperature at point T.



- (A) 18°C
 (B) 25°C
 (C) 20°C
 (D) 22°C

Ans. C

Sol.
$$\frac{(T-10)}{\left(\frac{3R}{2}\right)} + \frac{(T-15)}{\left(\frac{3R}{2}\right)} + \frac{(T-30)}{(R)} = 0$$

5. The electric field intensity in a region of space can be given as,
 $\vec{E} = c \frac{x\hat{i} + y\hat{j} + z\hat{k}}{(x^2 + y^2 + z^2)}$, where c is a constant of appropriate dimension. Find the net charge enclosed by a sphere of radius 'a' centred at origin.

- (A) $(4\pi\epsilon_0)c$
 (B) $(4\pi\epsilon_0)\left(\frac{c}{a}\right)$
 (C) $(4\pi\epsilon_0)ca$
 (D) $(4\pi\epsilon_0)\left(\frac{c}{a^2}\right)$

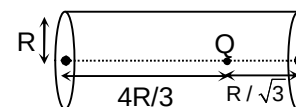
Ans. C

Sol. Taking the sphere as the gaussian surface

$$\phi_E = \frac{c}{a}(4\pi a^2) = \frac{q}{\epsilon_0}$$

$$\therefore q = (4\pi\epsilon_0)ca$$

6. A charged particle is placed on the axis of a cylinder as shown in the figure. Then, find the electric flux through the curved surface of the cylinder.



(A) $\left(\frac{7}{20}\right)\left(\frac{Q}{\epsilon_0}\right)$

(B) $\left(\frac{13}{20}\right)\left(\frac{Q}{\epsilon_0}\right)$

(C) $\left(\frac{13}{40}\right)\left(\frac{Q}{\epsilon_0}\right)$

(D) $\left(\frac{27}{40}\right)\left(\frac{Q}{\epsilon_0}\right)$

Ans. B

Sol. Solid angle subtended by the two plane surface are,

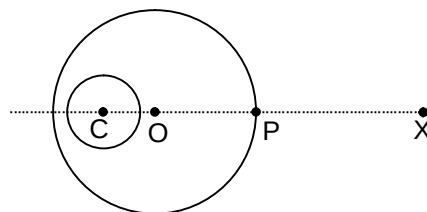
$$\Omega_{\text{plane}} = 2\pi(1 - \cos 60^\circ) + 2\pi(1 - \cos 37^\circ) = 2\pi\left(\frac{1}{2} + \frac{1}{5}\right) = \frac{7}{5}\pi$$

$\therefore$ Solid angle subtended by the curved surface is

$$\Omega_{\text{cs}} = 4\pi - \frac{7}{5}\pi = \frac{13}{5}\pi$$

$$\therefore \phi_{\text{cs}} = \frac{13\pi/5}{4\pi} \times \frac{Q}{\epsilon_0} = \left(\frac{13}{20}\right)\left(\frac{Q}{\epsilon_0}\right)$$

7. In an uncharged conducting sphere of radius R , a spherical cavity of radius $R/3$ is made as shown in the figure. It is given that $OC = \frac{R}{2}$ and $PX = R$. A point charge, q is placed at c and another point charge Q is placed at x . Find the potential of point P due to charges at the outer surface of the sphere.



(A) $\frac{k(Q - q)}{2R}$

(B) $\frac{k(Q - 2q)}{R}$

(C) $\frac{k(q - Q)}{R}$

(D) $\frac{k(2q - Q)}{2R}$

Ans. D

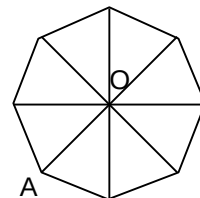
Sol. $V_P = V_0 = \frac{kQ}{2R} + \frac{kq}{R} = \frac{k(Q + 2q)}{2R}$

Potential of P due to 'Q' is $V_1 = \frac{kQ}{R}$

$\therefore$ Potential of P due to surface charge,

$$V = V_P - V_1 = \frac{k(2q - Q)}{2R}$$

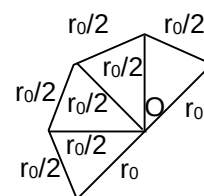
8. In the given diagram, each branch has a resistance r_0 . Then, the equivalent resistance between the points A and O is:



- (A) $\left(\frac{49}{105}\right)r_0$
 (B) $\left(\frac{51}{105}\right)r_0$
 (C) $\left(\frac{47}{105}\right)r_0$
 (D) $\left(\frac{53}{105}\right)r_0$

Ans. C

Sol. With the help of symmetry, the circuit can be simplified as:
 Then the circuit can be solved as simple series parallel circuit.



9. A galvanometer of resistance $20\ \Omega$ shows a deflection of 10 divisions when a current of 1 mA is passed through it. If a shunt of $4\ \Omega$ is connected and there are 50 divisions on the scale, then the range of the ammeter is:

- (A) 0.3 A
 (B) 1A
 (C) 3A
 (D) 30 mA

Ans. D

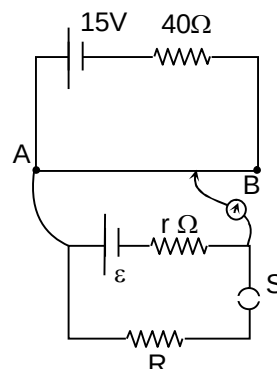
Sol. $I_g = 1\text{mA} \left(\frac{50}{10}\right) = 5\text{ mA}$

$$R_g = 20\Omega$$

$$I_g R_g = I_s R_s \Rightarrow I_s = 25\text{mA}$$

$$\therefore \text{Range of the ammeter } \varepsilon_r = I_g + I_s = 30\text{ mA}$$

10. In the shown potentiometer setup, length of AB is 10 m and has a resistance 20Ω . If the switch in the secondary circuit is open, the galvanometer gives zero deflection at 240 cm from point A. If the switch is closed and $R = x\Omega$, the zero deflection is at 180 cm from point A. If the switch is closed and $R = \frac{x}{3}\Omega$, the zero deflection is at y cm from point A. Then the value of y is



- (A) 90
(B) 120
(C) 60
(D) 80

Ans. B

Sol. Potential drop per unit length across

$$AB = \frac{\frac{1}{4} \times 20}{10} = \left(\frac{1}{2}\right) \text{ V/m}$$

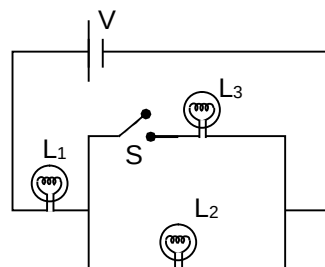
$$\therefore \varepsilon = \frac{1}{2} \times 2.4 \text{ V} = 1.2 \text{ V}$$

$$\text{Also, } \frac{\varepsilon x}{r+x} = \frac{1}{2} \times 1.8 \quad \dots(i)$$

$$\frac{\varepsilon \left(\frac{x}{3}\right)}{r + \left(\frac{x}{3}\right)} = \frac{1}{2} \times 1.2 \quad \dots(ii)$$

Hence, $y = 120\text{ cm}$

11. If L_1 , L_2 and L_3 in the figure are identical lamps, which of the following changes to the brightness of the lamps, occur when the switch S is closed?



- (A) In L_1 stays the same, in L_2 decreases
(B) In L_1 increases, in L_2 decreases

- (C) In L_1 increases, in L_2 stays the same
 (D) In L_1 decreases, in L_2 increases

Ans. B

Sol. Basic concept of current and voltage in series parallel circuit.

12. A thin rod of length $2\sqrt{3}R$ is charged uniformly with linear charge density λ . Another rod, semicircular in shape and radius R is also charged with the same linear charge density λ . Now, the rod is placed perpendicular to the plane of the semi-circular ring in such a way that the mid point of the rod coincides with the centre of the semi-circular ring. Find the magnitude of the force experienced by the rod.

- (A) $2\sqrt{3}k\lambda^2$
 (B) $\sqrt{3}k\lambda^2$
 (C) $\frac{\sqrt{3}}{2}k\lambda^2$
 (D) $k\lambda^2$

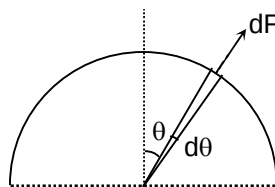
Ans. A

Sol. Electric field due to the rod at the ring,

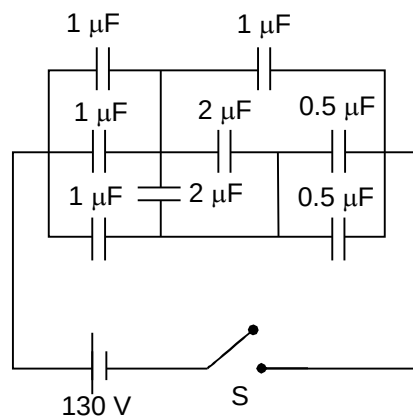
$$E = \frac{k\lambda}{R} \times 2 \times \sin 60^\circ = \frac{\sqrt{3}k\lambda}{R}$$

$$dF = \left(\frac{\sqrt{3}k\lambda}{R} \right) (\lambda R d\theta) = \sqrt{3}k\lambda^2 d\theta$$

$$F_{\text{net}} = 2 \int_0^{90^\circ} (dF) \cos \theta = 2\sqrt{3}k\lambda^2$$



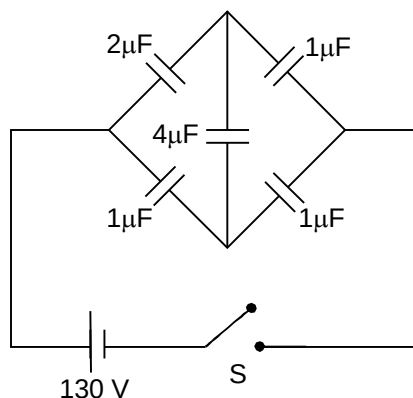
13. In the given capacitance circuit, all the capacitors are uncharged. Find the charge flown through the battery, as the switch 'S' is closed.



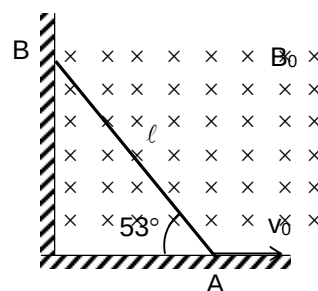
- (A) $135 \mu\text{C}$
 (B) $150 \mu\text{C}$
 (C) $155 \mu\text{C}$
 (D) $160 \mu\text{C}$

Ans. C

Sol. $C_{eq} = \frac{155}{130} \mu F$
 $\therefore Q = C_{eq} \times (130 \text{ V}) = 155 \mu C$



14. The end points of a uniform conducting rod slide on the two mutually perpendicular non-conducting surfaces in a uniform magnetic field as shown. If velocity of end A is V_0 and the rod makes an angle of 53° with the horizontal as shown. Then the maximum value of $|V_A - V_P|$ is (where P is a point on the rod) (V_A and V_P are potential of point A and P respectively).



- (A) $\frac{9}{125} B_0 \ell v_0$
 (B) $\frac{18}{125} B_0 \ell v_0$
 (C) $\frac{16}{125} B_0 \ell v_0$
 (D) $\frac{32}{125} B_0 \ell v_0$

Ans. D

Sol. $V_A - V_P$ is maximum when OP is perpendicular to AB

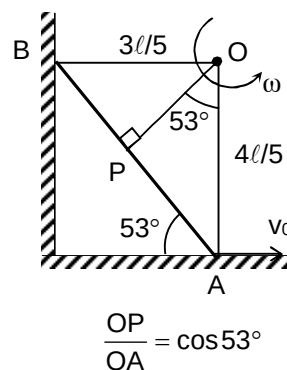
$$\omega = \frac{v_0}{4\ell/5} = \frac{5v_0}{4\ell}$$

$$|V_A - V_P|_{\max} = \frac{B\omega}{2} [(OA)^2 - (OP)^2]$$

$$= \frac{B_0}{2} \frac{5v_0}{4\ell} \left[\left(\frac{4\ell}{5} \right)^2 - \left(\frac{12\ell}{25} \right)^2 \right]$$

$$OP = \frac{3}{5} \left(\frac{4\ell}{5} \right) = \frac{12\ell}{25}$$

$$= \frac{5B_0 v_0 \ell^2 (20+12)(20-12)}{8\ell(25)^2} = \frac{32}{125} B_0 \ell v_0$$



15. A thermodynamic process consists of two isobars and two isotherms. The minimum and maximum temperatures are 400 K and 800 K. The maximum volume is 4 times the minimum volume and cycle does positive work and the gas contained by system is an ideal diatomic gas. The efficiency of the cyclic process is

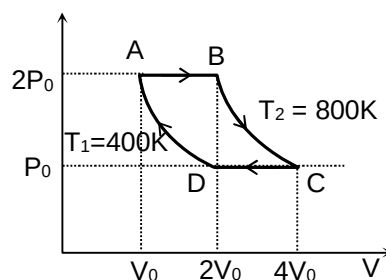
- (A) $\frac{2\ell n2}{7 - 4\ell n2}$
 (B) $\frac{2\ell n2}{7 - 2\ell n2}$
 (C) $\frac{2\ell n2}{7 + 4\ell n2}$
 (D) $\frac{2\ell n2}{7 + 2\ell n2}$

Ans. C

Sol. P-V curve for the given process is shown

$$\text{Also, } 2P_0V_0 = nR400$$

$$\eta = \frac{W_{\text{cycle}}}{Q^+} = \frac{2\ell n2}{7 + 4\ell n2}$$



16. One gram vapour at 100°C is added in a calorimeter which contains 20 gm mixture (of water and ice in equilibrium) to just melt the ice completely. Then the mass of water present in the mixture before the process is (Given: $S_w = 1 \text{ cal gm}^{-1}\text{C}^{-1}$, $L_f = 80 \text{ cal gm}^{-1}$, $L_v = 540 \text{ cal gm}^{-1}$)

- (A) 4 gm
 (B) 8 gm
 (C) 12 gm
 (D) 16 gm

Ans. C

Sol. Heat loss by vapour = Heat gain by ice

$$m_v L_v + m_v S_w (100 - 0) = m_{\text{ice}} L_f$$

$$1[540 + 100] = m_{\text{ice}} 80$$

$$m_{\text{ice}} = 8 \text{ gm}$$

$$m_{\text{water}} = 12 \text{ gm}$$

17. The magnetic needle performs 20 oscillations per minute in a horizontal plane. If the angle of dip is 30° , then how many oscillations per minute will this needle perform in vertical north-south plane

- (A) $20 \left[\frac{2}{3} \right]^{1/4}$
 (B) $20 \left[\frac{3}{2} \right]^{1/4}$
 (C) $20 \left[\frac{4}{3} \right]^{1/2}$
 (D) $20 \left[\frac{4}{3} \right]^{1/4}$

Ans. D

Sol. In horizontal plane, the magnetic needle oscillates in horizontal component

$$B_H, T_1 = 2\pi \sqrt{\frac{I}{MB_H}}$$

In vertical north south plane, the needle oscillates in the total earth's magnetic field B

$$T_2 = 2\pi \sqrt{\frac{I}{MB}}$$

$$\therefore \frac{T_1^2}{T_2^2} = \frac{B}{B_H} = (\sec \theta)^1 = \frac{2}{\sqrt{3}}$$

$$\frac{n_1^2}{n_2^2} = \left[\frac{3}{4} \right]^{1/2}$$

$$n_2^2 = \frac{2}{\sqrt{3}} n_1^2$$

$$n_2 = 20 \left[\frac{4}{3} \right]^{1/4}$$

18. A wooden cylinder of 50 kg is floating in water. A weight of 2 kg is to be placed on it to just make it submerged when the temperature is 10°C . How much less weight is to be placed when the temperature increases to 30°C . ($\gamma_{\text{water}} = 10^{-4}^\circ\text{C}^{-1}$, γ_{wood} is negligible)

- (A) 104 gm
 (B) 58 gm
 (C) 208 gm
 (D) 54 gm

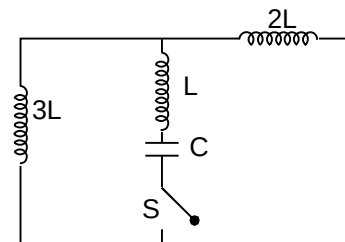
Ans. A

Sol. $(50 + 2)g = \rho_l Vg$

$$(50 + 2 - m)g = \frac{\rho_l Vg}{[1 + \gamma_l(30 - 10)]} \approx \rho_l [1 - \gamma_l(20)] Vg$$

$$\therefore m = 104 \text{ gm}$$

19. Initially there is no current in the inductors and capacitor is charged to a potential difference V_0 . Then the maximum energy stored in the inductor (L) after the switch is closed.



- (A) $\frac{5}{11}CV_0^2$
 (B) $\frac{5}{22}CV_0^2$
 (C) $\frac{5}{33}CV_0^2$
 (D) $\frac{5}{44}CV_0^2$

Ans. B

Sol. $i_1 + i_2 = i$... (i)

$$2L \frac{di_1}{dt} = 3L \frac{di_2}{dt}$$

$$2i_1 = 3i_2$$
 ... (ii)

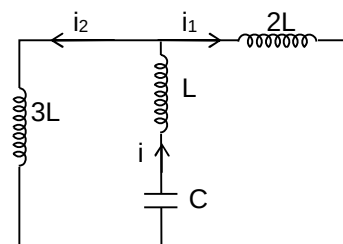
Solving (i) and (ii) $i_1 = \frac{3i}{5}$ and $i_2 = \frac{2i}{5}$

When i is maximum then i_1 and i_2 are also maximum, hence the charge on capacitor is zero at this instant.

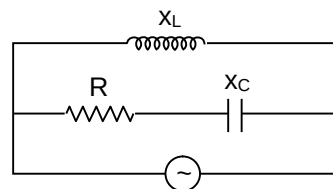
Applying conservation of Energy

$$\frac{CV_0^2}{2} = \frac{Li^2}{2} + \frac{2L}{2} \left(\frac{3i}{5} \right)^2 + \frac{3L}{2} \left(\frac{2i}{5} \right)^2$$

$$i^2 = \frac{5CV_0^2}{11L}, \quad U_{\max}(\text{in } L) = \frac{5CV_0^2}{22}$$



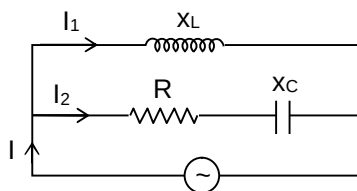
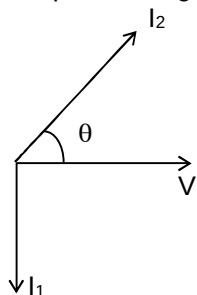
20. If the wattless current through the AC source is zero in the given circuit. Then which of the following is correct.



- (A) $R^2 = x_L(x_C - x_L)$
 (B) $R^2 = x_C(x_L - x_C)$
 (C) $R^2 = x_C(x_L + x_C)$
 (D) $R^2 = x_L(x_C + x_L)$

Ans. B

Sol. The phasor diagram for the following circuit is shown as



$$\cos \theta = \frac{R}{\sqrt{R^2 + X_C^2}}$$

If wattless current is zero, then I and V must be in the same phase,

$$\therefore I_1 = I_2 \sin \theta$$

$$\frac{V}{X_L} = \frac{V}{\sqrt{X_C^2 + R^2}} \frac{X_C}{\sqrt{X_C^2 + R^2}}$$

$$X_C^2 + R^2 = X_C X_L$$

$$R^2 = X_C (X_L - X_C)$$

SECTION – B (Numerical Answer Type)

This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE**. For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

21. Two uniform spheres A and B are made up of the same material which is a good conductor of heat. The radius of A is twice the radius of B. The rate of fall of temperature of A is 'x' times that of the B when both are at the same surface temperature. The value of 'x' is

Ans. 00000.50

Sol. $\frac{dQ}{dt} \propto R^2$

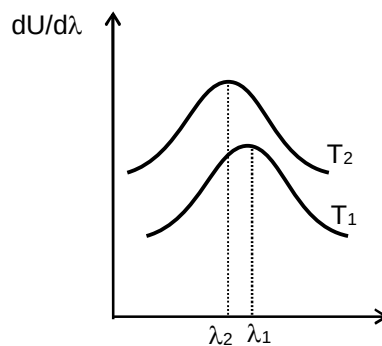
Heat capacity $H \propto R^3$

Clearly, $\frac{dT}{dt} = \frac{\left(\frac{dQ}{dt}\right)}{H}$

$$\therefore \frac{dT}{dt} \propto \left(\frac{1}{R}\right)$$

$$\therefore \frac{\left(\frac{dT}{dt}\right)_A}{\left(\frac{dT}{dt}\right)_B} = \frac{R_B}{R_A} = \frac{1}{2}$$

22. The spectral emissive power for a body at temperature T_1 and T_2 are plotted against the wavelength. The area under the curve at temperature T_2 is four times the area under the curve at temperature T_1 . Then, λ_1/λ_2 is equal to:



Ans. 00001.41

Sol. Area under the curve,
 $U \propto T^4$

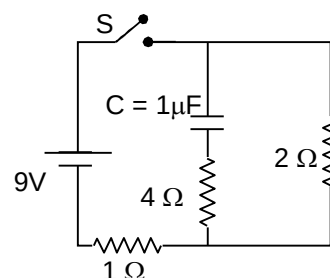
$$\therefore \frac{U_2}{U_1} = \left(\frac{T_2}{T_1}\right)^4 = 4$$

$$\Rightarrow \frac{T_2}{T_1} = \sqrt{2}$$

$$\text{Also, } \lambda_1 T_1 = \lambda_2 T_2$$

$$\Rightarrow \left(\frac{\lambda_1}{\lambda_2}\right) = \frac{T_2}{T_1} = \sqrt{2} = 1.41$$

23. In the given circuit, switch 'S' is closed at $t = 0$ and initially capacitor is uncharged. Then the charge (in μC) on the capacitor after long time is



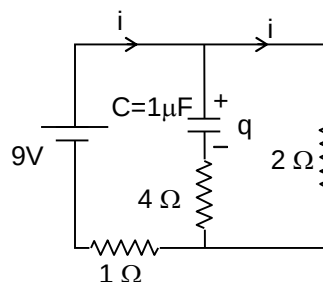
Ans. 00006.00

Sol. After long time current in the branch of the capacitor is zero.

$$\text{Therefore, } i = \frac{9}{1+2} = 3 \text{ A}$$

$$\frac{q}{C} = 2(3)$$

$$q = 6 \mu\text{C}$$



24. A bar magnet is inserted inside a solenoid of length 0.2 m and 100 turns. It gets completely magnetized if a current of 4A is passed through it. The coercivity of the bar magnet is $n \times 1000$ A/m. The value of 'n' is

Ans. 00002.00

Sol. $B = \mu_0 H = \mu_0 n I$
 $\Rightarrow H = n I = 2000 \text{ A/m}$

25. A closed vessel of volume V_0 contains oxygen gas at a pressure P_0 and temperature T_0 . Another closed vessel of same volume V_0 contains helium gas at pressure P_0 and temperature $\frac{T_0}{2}$. Find the ratio of the mass of helium gas to the oxygen gas.

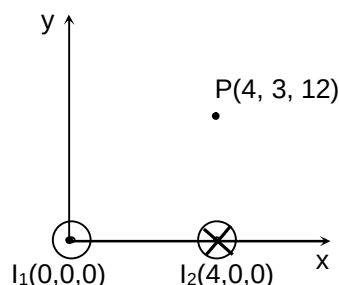
Ans. 00000.25

Sol. Using, $PV = nRT$ for both gases

$$P_0 V_0 = \frac{m_{\text{He}}}{M_{\text{He}}} R \frac{T_0}{2} \quad P_0 V_0 = \frac{m_{\text{O}_2}}{M_{\text{O}_2}} R T_0$$

$$\frac{m_{\text{He}}}{m_{\text{O}_2}} = \frac{2M_{\text{He}}}{M_{\text{O}_2}} = \frac{2(4)}{32} = \frac{1}{4} = 0.25$$

26. Two infinite current carrying wires having currents I_1 and I_2 are placed as shown. If point P has a net magnetic field along y-axis, then the ratio $\left(\frac{I_1}{I_2}\right)$ is



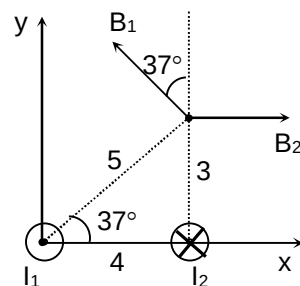
Ans. 00002.78

Sol. $B_x = 0$

$$B_2 = \frac{3B_1}{5}$$

$$\frac{\mu_0 I_2}{2\pi(3)} = \frac{3}{5} \frac{\mu_0 I_1}{2\pi(5)}$$

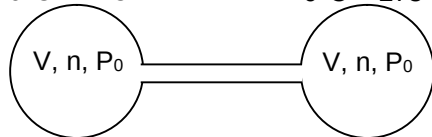
$$\frac{I_1}{I_2} = \frac{25}{9} = 2.78$$



27. Two glass bulbs of equal volume are connected by a narrow tube and are filled with a gas at 0°C and a pressure of 76 cm of mercury. One of the bulb is then placed in melting ice and the other is placed in water bath maintained at 62°C . The new value of pressure inside the bulb is (assuming that volume of connecting tube is negligible) x cm of mercury. Find the value of x .

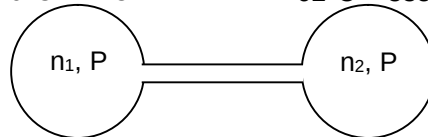
$0^\circ\text{C} = 273 \text{ K}$

$0^\circ\text{C} = 273 \text{ K}$



$0^\circ\text{C} = 273 \text{ K}$

$62^\circ\text{C} = 335 \text{ K}$



Ans. 00083.75

Sol. Total number of moles remains conserved as the whole system is closed.

$$n_1 + n_2 = 2n$$

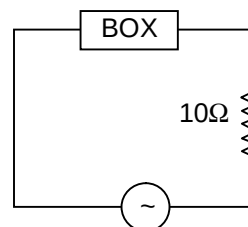
$$\frac{PV}{R(273)} + \frac{PV}{R(335)} = \frac{2P_0V}{R(273)}$$

$$P = \frac{335}{304}(P_0)$$

$$P = \frac{335}{304}(76 \text{ cm of Hg}) = 83.75 \text{ cm of Hg}$$

$$x = 83.75$$

28. In the circuit shown in figure, power factor of the box is 0.5 and the power factor of the circuit is given as $\frac{\sqrt{3}}{2}$. If the current is lagging the voltage. The effective resistance (in Ω) of the box is



Ans. 00005.00

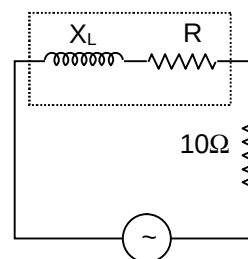
Sol. If current is lagging the voltage and power factor of the box is 0.5, then box must contain a resistance in series with an inductor. Now the circuit looks like

$$\frac{R^2}{x_L^2 + R^2} = (0.5)^2 \quad \dots(i)$$

$$\frac{(R+10)^2}{x_L^2 + (R+10)^2} = \left(\frac{\sqrt{3}}{2}\right)^2 \quad \dots(ii)$$

Solving (i) and (ii)

$$R = 5\Omega$$



29. A particle having charge q and mass m is projected with a velocity $(4\hat{i} - 6\hat{j} + 3\hat{k})$ m/s from the origin in a region occupied by electric field (E) and magnetic field (B) such that $\vec{E} = E_0\hat{j}$ and $\vec{B} = B_0\hat{j}$. Find the time (in sec) after which the speed of charged particle will be $5\sqrt{5}$ m/s (neglect the gravity) (take $\frac{qE_0}{m} = 2 \text{ m/s}^2$)

Ans. 00008.00

Sol. $v_y = u_y + a_y t$

$$0 = 6 - \frac{qE_0}{m} t_1 \quad t_1 = 3 \text{ sec}$$

$$v_x^2 + v_y^2 + v_z^2 = 125 \quad v_y = 10 \text{ m/s}$$

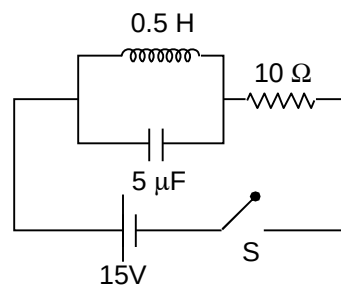
$$(4)^2 + v_y^2 + (3)^2 = 125$$

$$v_y = u_y + a_y t$$

$$10 = 0 + \frac{qE_0}{m} t_2 \quad t_2 = 5 \text{ sec}$$

$$t = t_1 + t_2 = 8 \text{ sec}$$

30. In the circuit shown, the current at $t = 0$ is I_1 and the current at time $t \rightarrow \infty$ is I_2 . Then the value of $(I_1 + I_2)$ in ampere is



Ans. 00003.00

Sol. $I_1 = 1.5 \text{ A}$
 $I_2 = 1.5 \text{ A}$
 $I_1 + I_2 = 3 \text{ A}$

Chemistry

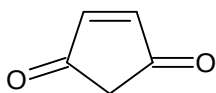
PART – B

SECTION – A (One Options Correct Type)

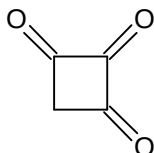
This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

31. Which of the following compound has negligible enol content?

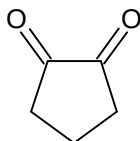
(A)



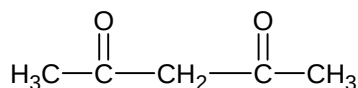
(B)



(C)



(D)

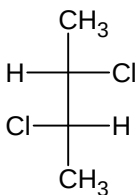


Ans. A

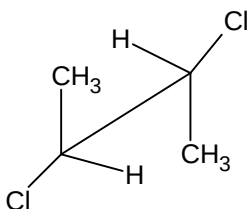
Sol. Option (A) has negligible enol content because its enol would be highly unstable (Anti – Aromatic).

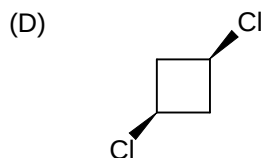
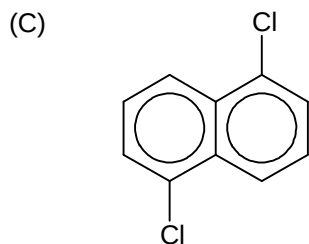
32. Which of the following is a meso compound

(A)



(B)

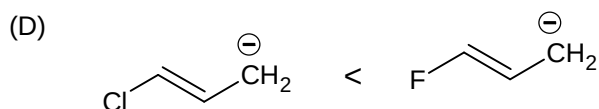
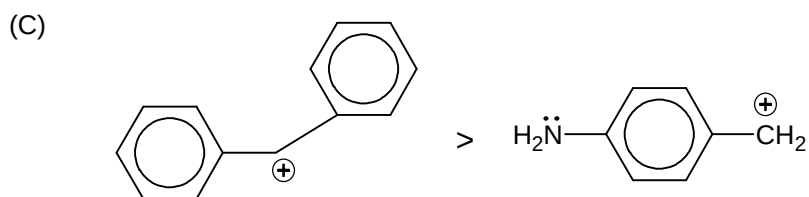
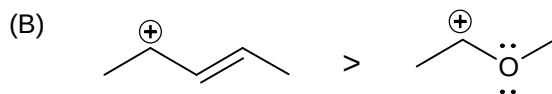
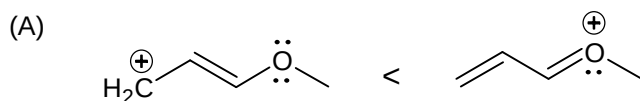




Ans. B

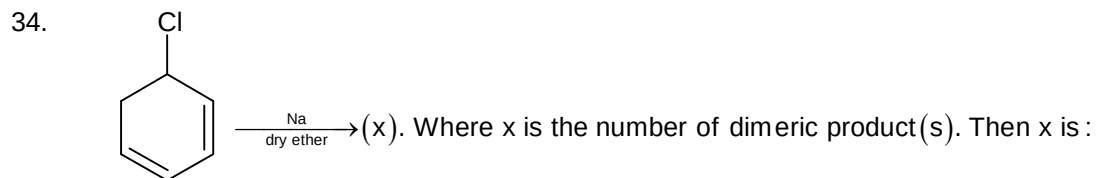
Sol. Meso compound are those compounds which contain chiral carbon but are optically inactive due to presence of POS or COS or both.

33. Which of the given following stability order is correct?



Ans. A

Sol. Option (A) is correct because second resonating structure has complete octet of all its atoms.



(A) 1

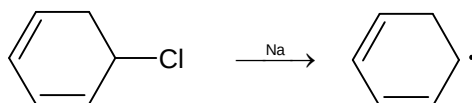
(B) 2

(C) 3

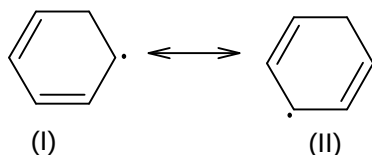
(D) 4

Ans. C

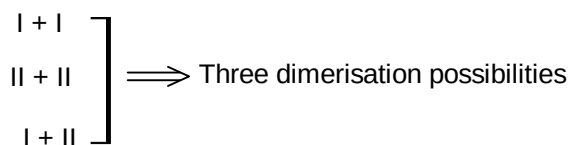
Sol. It's Wurtz reaction which involves formation of free-radical.



This free radical has resonating structures.



Dimerisation of these free radicals can form three product as there are three possibilities:



35. Which one of the following is most reactive for electrophilic addition reactions?

 (A) $\text{CH}_3 - \text{CH} = \text{CH}_2$

 (B) $\text{CH}_3 - \text{O} - \text{CH} = \text{CH}_2$

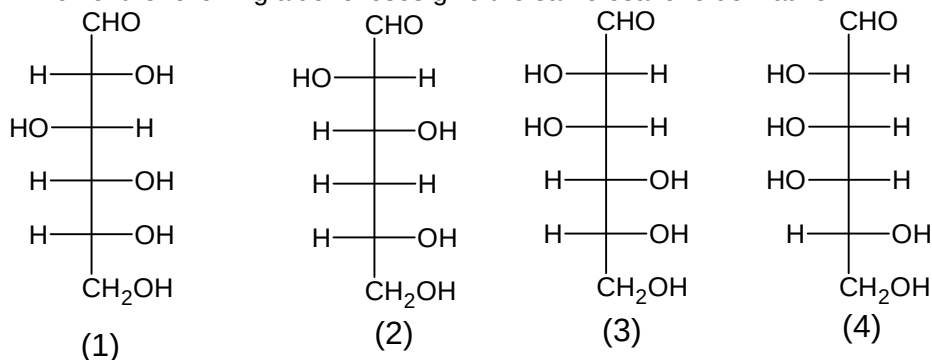
 (C) $\text{C}_6\text{H}_5 - \text{CH} = \text{CH}_2$

 (D) $\text{CH}_3 - \text{CH} = \text{CH} - \text{CH}_3$

Ans. B

Sol. Option (B) is most reactive because carbocation from option (B) would be most stable because its resonating structure will have complete octet of all its atoms.

36. Which of the following aldohexoses give the same osazone derivative?



- (A) 1 and 4
 (B) 1 and 3
 (C) 2 and 3
 (D) 3 and 4

Ans. B

Sol. 1 and 3 has identical configuration at carbon number 3, 4 and 5. Hence, give same osazone.

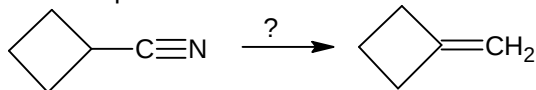
37. You have a mixture of three amino acids E(PI = 3.2), Y(PI = 5.7) and K(PI = 9.7). Under electrophoresis at pH = 7.7, in which direction will each component of the mixture move? (PI stands for isoelectric point of amino acid)

- (A) E towards anode side; Y and K towards cathode side
 (B) E towards anode side; Y constants; K towards cathode side
 (C) E towards cathode side; Y constant; K towards anode side
 (D) E and Y towards anode side; K towards cathode side

Ans. D

Sol. An amino acid will be positively charged if the pH of the solution is less than PI (iso-electric point) of the amino acid, it will be negatively charged if the pH of the solution is greater than the PI of the amino acid.
 Therefore, option **(D)** is correct.

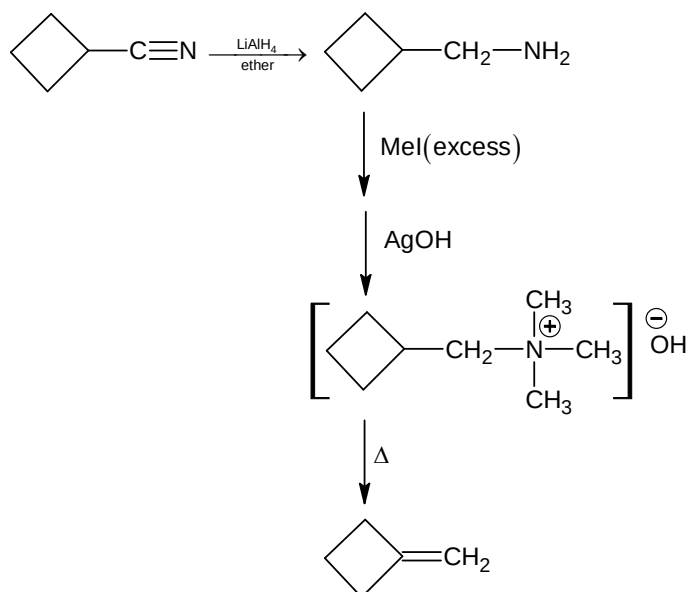
38. What sequence of reactions would best accomplish the following reaction?



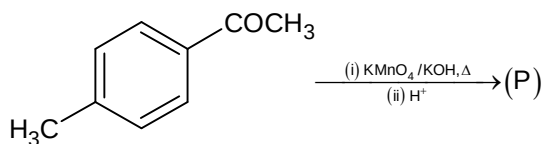
- (A) (i) LiAlH_4 in ether; (ii) $3\text{CH}_3\text{I}$ followed by heating with AgOH
 (B) (i) LiAlH_4 in ether; (ii) P_2O_5 and heat
 (C) (i) 20% H_2SO_4 and heat; (ii) P_2O_5 and heat
 (D) (i) H_2 and Lindlar catalyst

Ans. A

Sol.



39.

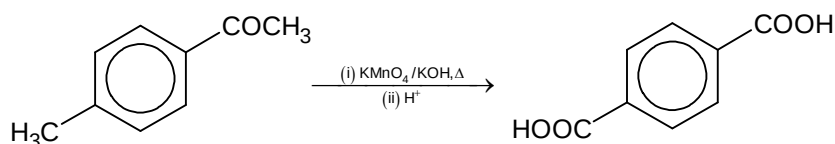


The product (P) is:

- (A)
- (B)
- (C)
- (D)

Ans. C

Sol.



40. CH_3COOH $\text{H}-\text{COOCH}_3$ $\text{CH}_3-\text{CONH}_2$ $\text{CH}_3-\text{CH}_2-\text{CH}_2-\text{OH}$
 (I) (II) (III) (IV)

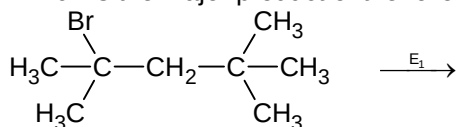
The correct increasing order of boiling points is

- (A) $\text{II} < \text{IV} < \text{I} < \text{III}$
 (B) $\text{I} < \text{II} < \text{III} < \text{IV}$
 (C) $\text{IV} < \text{II} < \text{I} < \text{III}$
 (D) $\text{III} < \text{I} < \text{IV} < \text{II}$

Ans. A

Sol. B.P. $\propto$ extent of inter-molecular H-bonding.

41. Which is the major product of the following elimination reaction?

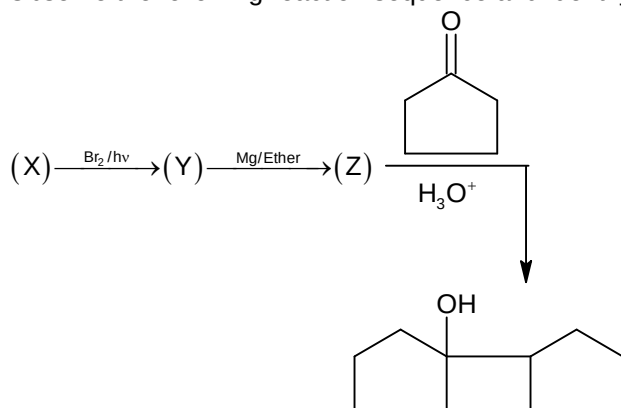


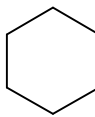
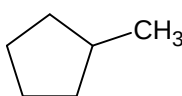
- (A) $\begin{array}{c} \text{H}_3\text{C} \\ | \\ \text{H}_3\text{C}-\text{C}-\text{CH}-\text{CH}=\text{CH}_2 \\ | \quad | \\ \text{H}_3\text{C} \quad \text{CH}_2 \end{array}$
 (B) $\begin{array}{c} \quad \quad \text{CH}_3 \\ \quad \quad | \\ \text{H}_3\text{C}-\text{C}=\text{C}-\text{CH}-\text{CH}_3 \\ | \quad \quad | \\ \text{H}_3\text{C} \quad \text{CH}_3 \end{array}$
 (C) $\begin{array}{c} \quad \quad \text{H} \\ \quad \quad | \\ \text{H}_3\text{C}-\text{C}=\text{C}-\text{C}-\text{CH}_3 \\ | \quad \quad | \\ \text{H}_3\text{C} \quad \text{CH}_3 \end{array}$
 (D) $\begin{array}{c} \quad \quad \text{CH}_3 \\ \quad \quad | \\ \text{H}_2\text{C}=\text{C}-\text{CH}_2-\text{C}-\text{CH}_3 \\ \quad \quad | \\ \quad \quad \text{CH}_3 \end{array}$

Ans. D

Sol. Due to steric hindrance, Hoffmann product is the major product.

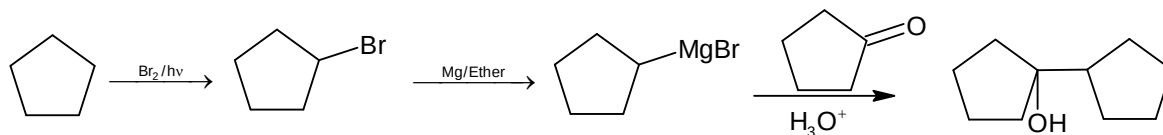
42. Observe the following reaction sequence and identify (X).



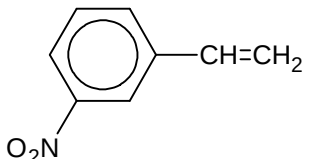
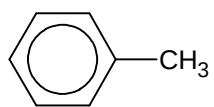
- (A) 
- (B) 
- (C) 
- (D) 

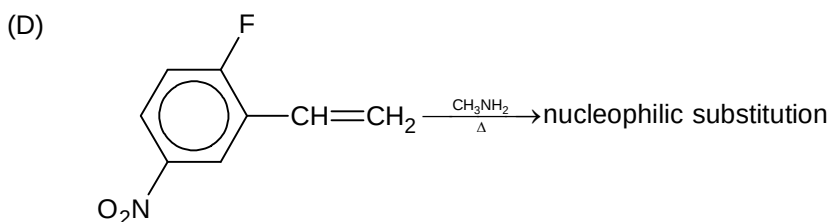
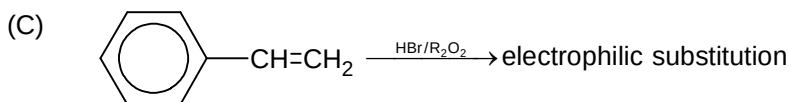
Ans. A

Sol.



43. Which of the following is correctly matched?

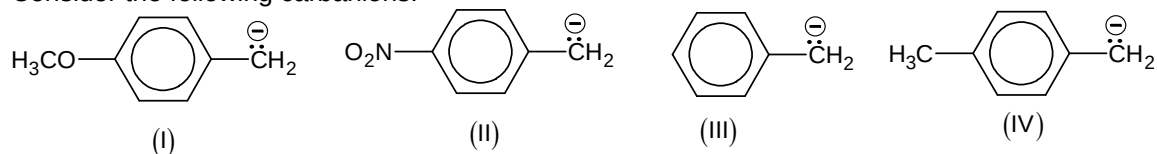
- (A)  $\xrightarrow{Br_2/Fe}$ free radical reaction
- (B)  $\xrightarrow{Cl_2/Fe}$ electrophilic addition



Ans. D

Sol. Factual question.

44. Consider the following carbanions:



The correct decreasing order of pK_b is:

- (A) II > III > IV > I
 (B) III > IV > I > II
 (C) I > IV > III > II
 (D) II > I > III > IV

Ans. A

Sol. Carbanion are Lewis bases. Electron withdrawing groups decrease the basic nature and electron donating groups increase the basic nature and as we know that more the basic nature, lesser the pK_b .

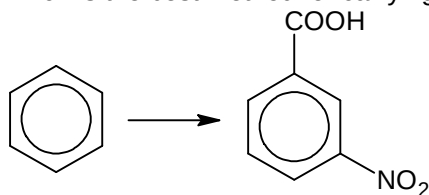
45. Which of the following polymer, not having C – C bond formation during polymerization?

- (A) Teflon
 (B) Bakelite
 (C) Natural rubber
 (D) Terylene

Ans. D

Sol. Refer to NCERT.

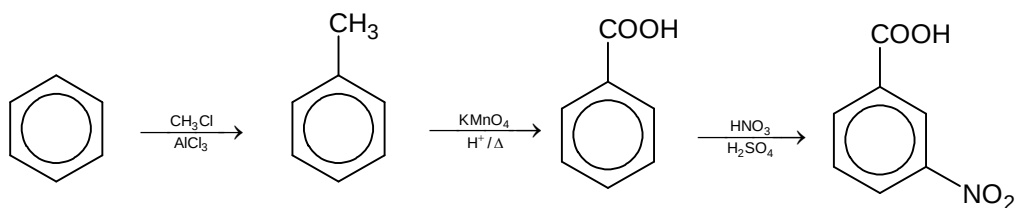
46. Which is the best method for carrying out the following reaction?



- (A) $\text{CH}_3\text{Cl} / \text{AlCl}_3; \text{KMnO}_4 / \text{H}^+; \text{heat}; \text{HNO}_3 / \text{H}_2\text{SO}_4$
 (B) $\text{CH}_3\text{Cl} / \text{AlCl}_3; \text{HNO}_3 / \text{H}_2\text{SO}_4; \text{KMnO}_4 / \text{H}^+; \text{heat}$
 (C) $\text{HNO}_3 / \text{H}_2\text{SO}_4; \text{CH}_3\text{Cl} / \text{AlCl}_3; \text{KMnO}_4 / \text{H}^+; \text{heat}$
 (D) $\text{HNO}_3 / \text{H}_2\text{SO}_4; \text{Sn} / \text{HCl}; \text{H}_3\text{O}^+$

Ans. A

Sol.



47. The compound which does not give blue colour in Lassaigne's test is:

- (A) Aniline
 (B) Glycine
 (C) Hydrazine
 (D) Urea

Ans. C

Sol. Lassaigne's test can be used for detection of N in organic compounds.

48. Drugs which do not bind to the enzyme's active site, but to different site. This different site is

- (A) Inhibition site
 (B) Competitive site
 (C) Allosteric site
 (D) None of these

Ans. C

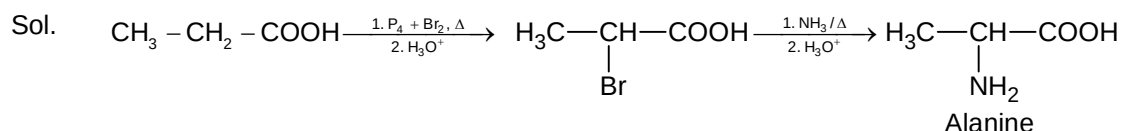
Sol. Refer to NCERT.

49. What is the name of the amino acid produced when propanoic acid is subjected to the following sequence of reagents:

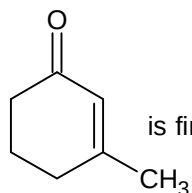
1. Red P + Br₂ / Δ
2. H₃O⁺
3. NH₃, Δ
4. H₃O⁺

- (A) Alanine
(B) Aspartic acid
(C) Asparagine
(D) Glutamic acid

Ans. A



50.

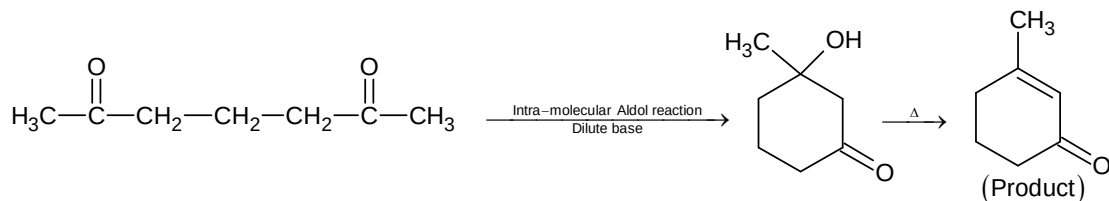


is final product when which of following is reacted with base followed by heating?

- (A) $\text{H}_3\text{C} - \overset{\text{O}}{\parallel} \text{C} - \text{CH}_2 - \text{CH}_2 - \text{CH}_2 - \overset{\text{O}}{\parallel} \text{C} - \text{CH}_3$
(B) $\text{H}_3\text{C} - \text{CH}_2 - \text{CH}_2 - \overset{\text{O}}{\parallel} \text{C} - \text{CH}_2 - \overset{\text{O}}{\parallel} \text{C} - \text{CH}_3$
(C) $\text{H}_3\text{C} - \overset{\text{O}}{\parallel} \text{C} - \text{CH}_2 - \text{CH}_2 - \text{CH}_2 - \text{CH}_2 - \overset{\text{O}}{\parallel} \text{C} - \text{CH}_2 - \text{Cl}$
(D) $\text{H}_3\text{C} - \overset{\text{O}}{\parallel} \text{C} - \text{CH}_2 - \text{CH} = \text{CH} - \overset{\text{O}}{\parallel} \text{C} - \text{CH}_3$

Ans. A

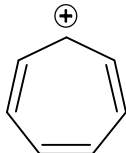
Sol.



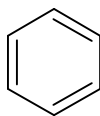
SECTION – B (Numerical Answer Type)

This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE**. For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

51. Consider the following structure



Tropylium cation



Benzene

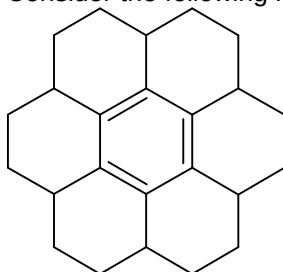
If x and y are the number of major resonating structures of tropylium cation and benzene, then what is the value of $\frac{x}{y}$?

Ans. 00003.50

Sol. There are seven resonating structure for tropylium cation and two resonating structures for benzene.

$$\left[\frac{7}{2} = 3.5 \right]$$

52. Consider the following molecule

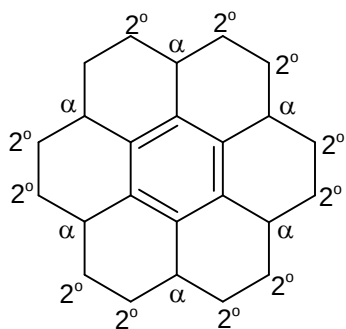


If x = number of α – hydrogen
 y = number of secondary hydrogen

What is the value of $\frac{x}{y}$?

Ans. 00000.25

Sol.



Number of α - H = 6

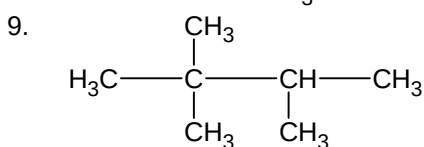
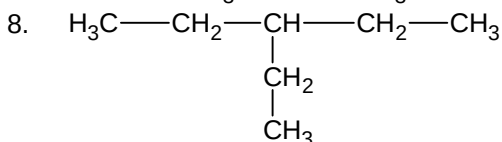
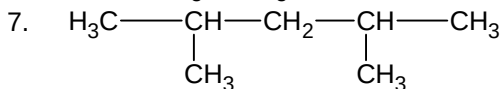
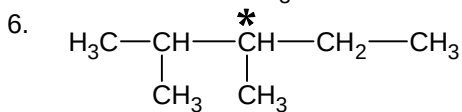
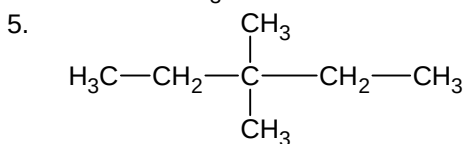
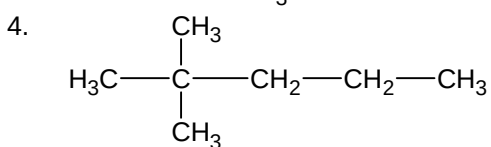
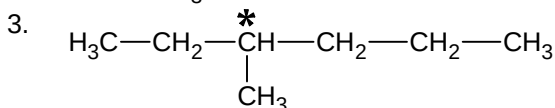
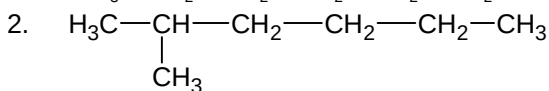
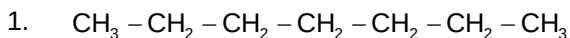
Number secondary hydrogen = 24

$$\left[\frac{6}{24} \right] = 0.25$$

53. If x are the total number of structural isomers for molecular formula C_7H_{16} and y are the number of isomers which contain chiral carbon. What is the value of $\frac{x}{y}$?

Ans. 00004.50

Sol. Total number of structural isomers for $C_7H_{16} = 9$



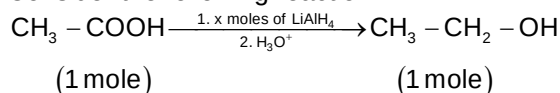
Out of these 9 isomers, only two isomers (3 and 6) contain chiral carbon.

Therefore, $x = 9$

$$y = 2$$

$$\frac{9}{2} = 4.5$$

54. Consider the following reaction



What is the minimum value of x ?

Ans. 00000.75

Sol. 1 mole of carboxylic acid requires at least 0.75 moles of LiAlH_4 for its reduction into alcohol.

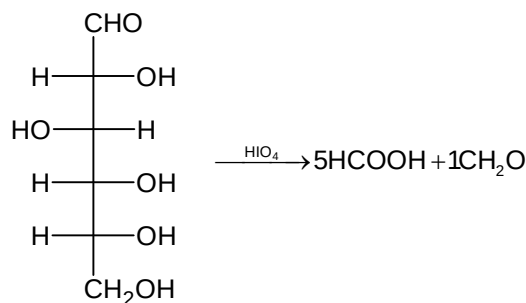
55. $\text{D-Glucose} \xrightarrow{\text{HIO}_4} x \text{ moles of HCOOH}$

$$\text{D-Fructose} \xrightarrow{\text{HIO}_4} y \text{ moles of HCOOH}$$

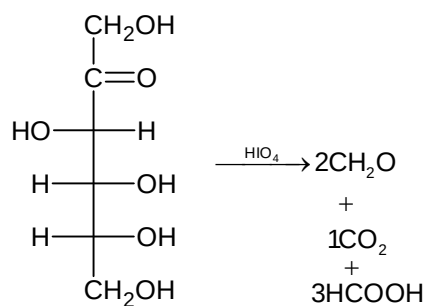
What is the value of $\frac{x}{y}$?

Ans. 00001.67

Sol.



D-Glucose



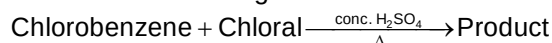
D-Fructose

$x = 5$

$$y = 3$$

$$\frac{5}{3} = 1.67$$

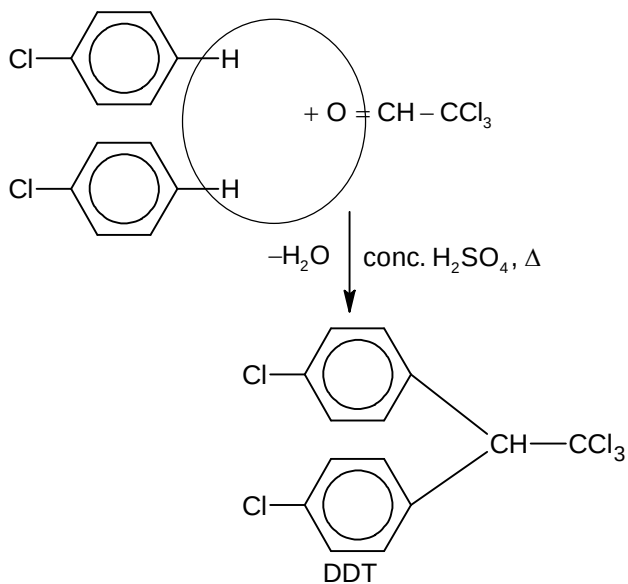
56. Consider the following reaction



What is % of chlorine in the product?

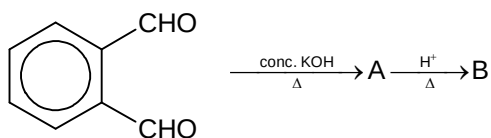
Ans. 00050.07

Sol.



$$\% \text{ of chlorine in DDT} = \frac{177.5}{354.5} \times 100 = 50.07\%$$

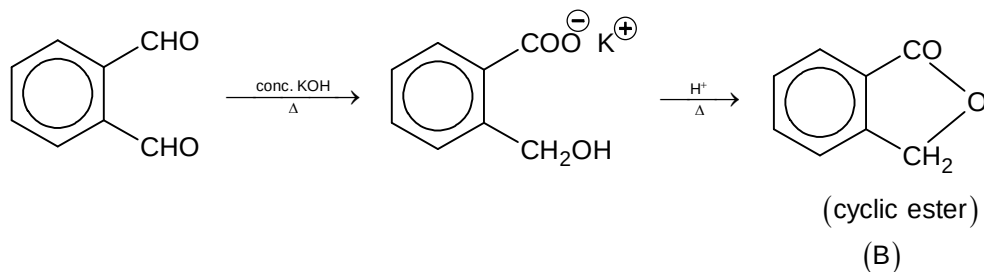
57.



What is the ratio of $\frac{\text{degree of unsaturation}}{\text{number of } sp^2 \text{ hybridized carbon}}$ in B?

Ans. 00000.86

Sol.



Degree of unsaturation of B = 6

Number of sp^2 hybridized carbon in B = 7

$$\frac{6}{7} = 0.86$$

58. Consider the following reaction

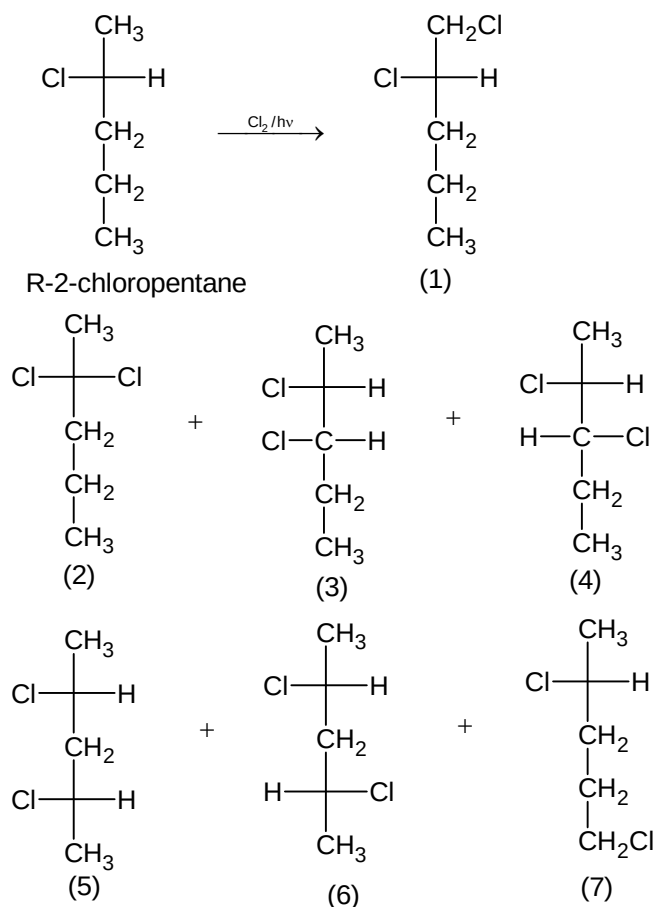
R - 2 - chloropentane $\xrightarrow[h\nu]{Cl_2}$ optically active di - chloro products (P)

 $\xrightarrow[h\nu]{Cl_2}$ optically active di - chloro products (Q)
R - 2 - chlorobutane $\xrightarrow[h\nu]{Cl_2}$ optically active di - chloro products (R)

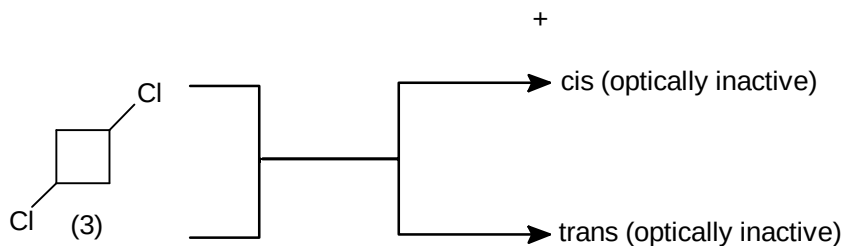
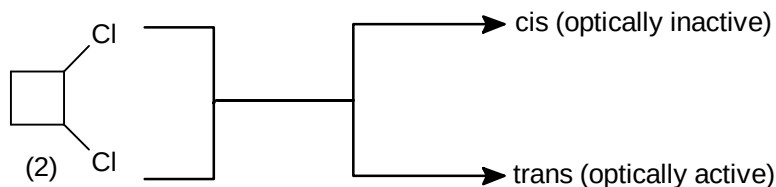
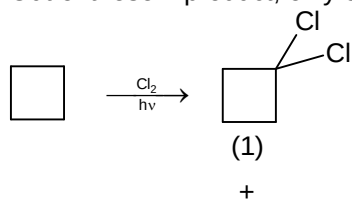
What is the value of $\frac{Q+R}{P}$?

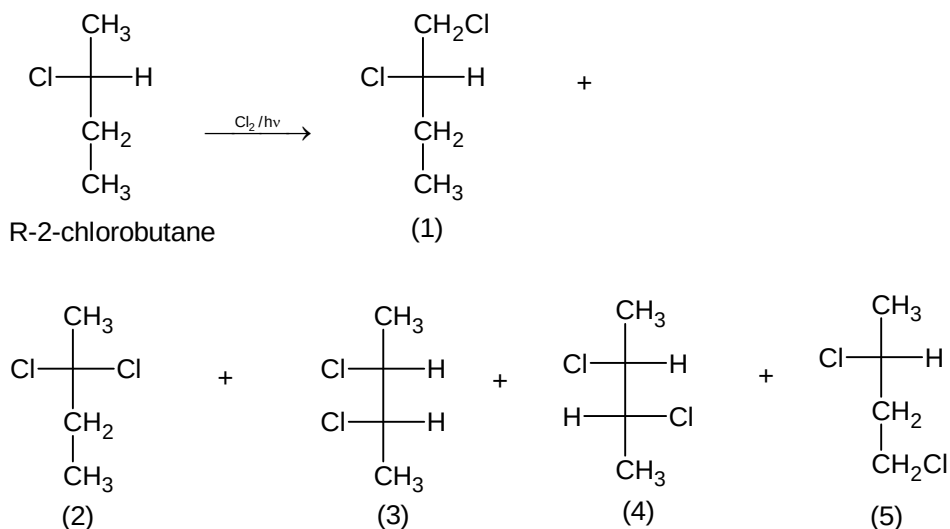
Ans. 00001.00

Sol.



Out of these 7 product, only 5 (1, 3, 4, 6 and 7) are optically active.


 $\therefore Q = 2$



Out of these 5 products, only three (1, 4 and 5) are optically active.

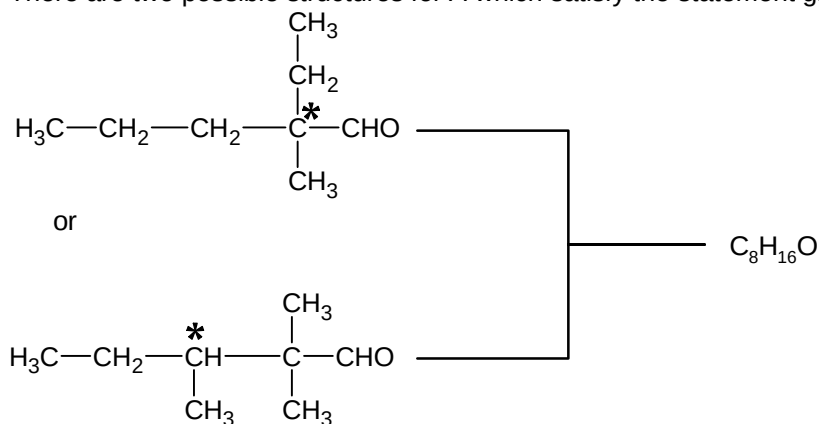
$\therefore R = 3$

Thus, $\frac{Q+R}{P} = \frac{2+3}{5} = 1.00$

59. A non-cyclic aldehyde A having molecular formula $\text{C}_x\text{H}_y\text{O}$ has one chiral carbon. A does not participate in Aldol condensation. Calculate ratio of number of possible structural isomers justifying above statement
minimum value of x

Ans. 00172.26

Sol. There are two possible structures for A which satisfy the statement given in the question



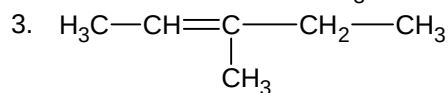
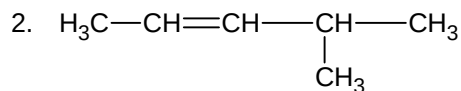
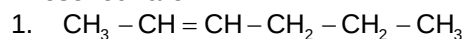
$\therefore \frac{2}{8} = \frac{1}{4} = 0.25$

60. If x is the total number of alkenes of molecular formula C_6H_{12} which show geometrical isomerism. If y is the total number of cyclic isomers for molecular formula C_5H_{10} . Then, what is the value of $\frac{y}{x}$?

Ans. 00001.75

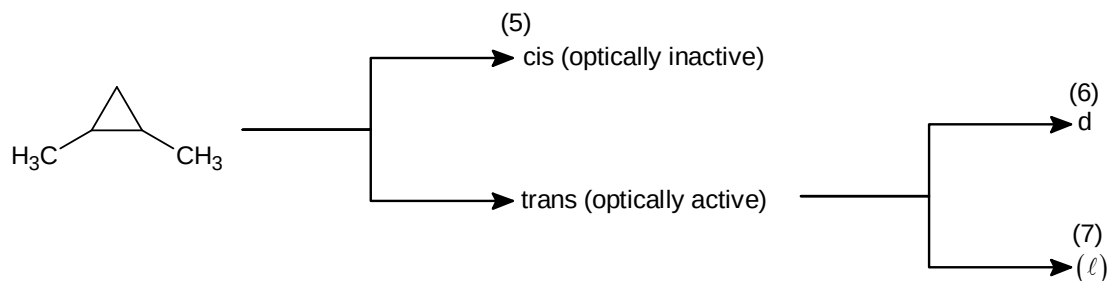
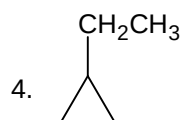
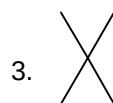
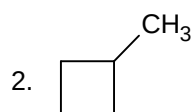
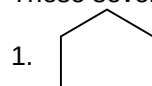
Sol. Total number of alkenes of MF C_6H_{12} showing geometrical isomerism = 4

These four are:



Total number of cyclic isomer of MF $C_5H_{10} = 7$

These seven are:



$$\therefore y = 7, x = 4$$

$$\therefore \frac{y}{x} = \frac{7}{4} = 1.75$$

Mathematics

PART – C

SECTION – A (One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

61. In a ΔABC $a^2 \sin 2B + b^2 \sin 2A$ is equal to (where Δ is area of ΔABC)

- (A) 4Δ
- (B) 2Δ
- (C) Δ
- (D) $\frac{\Delta}{2}$

Ans. A

Sol. $a^2 \sin 2B + b^2 \sin 2A = 4R^2(\sin^2 A(2 \sin B \cos B) + (\sin^2 B(2 \sin A \cos A)))$
 $\Rightarrow 8R^2 \sin A \sin B (\sin A \cos B + \cos B \sin A)$
 $\Rightarrow 8R^2 \sin A \sin B \sin C$
 $\Rightarrow 4\Delta$

62. If the distance between the foci of an ellipse is 8 and distance between its directrix is 10, then length of its latus rectum is

- (A) $\sqrt{5}$
- (B) $\frac{2}{\sqrt{5}}$
- (C) $\frac{4}{\sqrt{5}}$
- (D) $2\sqrt{5}$

Ans. C

Sol. $2ae = 8$ and $\frac{2a}{e} = 10$

$$\frac{2ae \cdot e}{2a} = \frac{8}{10} \Rightarrow e^2 = \frac{8}{10} \Rightarrow e = \sqrt{\frac{8}{10}} = \frac{2}{\sqrt{5}}$$

$$\text{Length of latus rectum} = \frac{2b^2}{a} = \frac{2 \cdot a^2(1-e^2)}{a} = 2a(1-e^2) = \frac{8\sqrt{5}}{2} \left(1 - \frac{8}{10}\right) = 4\sqrt{5} \left(\frac{2}{10}\right) = \frac{4}{\sqrt{5}}$$

63. The locus of the mid-points of the perpendiculars drawn from points on the line $x + y + 5 = 0$ to the line $2x - y + 3 = 0$ is

- (A) $2y + x - 6 = 0$
 (B) $11x - 8y - 16 = 0$
 (C) $11x + 8y + 16 = 0$
 (D) $y - x + 2 = 0$

Ans. B

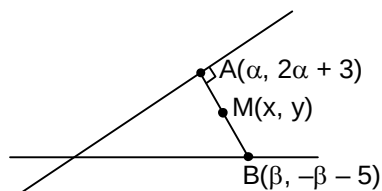
Sol. $\frac{y - (2\alpha + 3)}{x - \alpha} \times 2 = -1$
 $2(y - 2\alpha - 3) = \alpha - x$
 $2y - 4\alpha - 6 = \alpha - x$
 $2y - 6 + x = 5\alpha \dots (1)$
 $\frac{\beta + \alpha}{2} = x, \frac{-\beta - 5 + 2\alpha + 3}{2} = y$

$\beta + \alpha = 2x, 2y - 2\alpha + 2 + \alpha = 2x$

$2y - 2x + 2 = \alpha$

$-\beta + 2\alpha - 2 = 2y, \beta = 2y - 2\alpha + 2$

So, equation $2y - 6 + x = 5(2y - 2x + 2)$; $2y - 6 + x = 10y - 10x + 10$; $11x - 8y - 16 = 0$

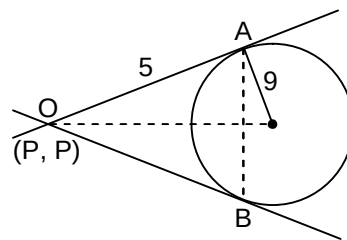


64. Let the tangents drawn from the point $(1, 1)$ to the circle $x^2 + y^2 + 16x + 8y - 1 = 0$ touch it at the points A and B. The $(AB)^2$ is equal to

- (A) $\frac{90}{106}$
 (B) $\frac{81}{106}$
 (C) $\frac{8100}{212}$
 (D) $\frac{4050}{53}$

Ans. D

Sol. $OA = \sqrt{5} = \sqrt{1+1+16+8-1} = 5$
 Length of $AB = \frac{2LR}{\sqrt{L^2 + R^2}} = \frac{2 \times 5 \times 9}{\sqrt{25 + 81}} = \frac{90}{\sqrt{106}}$
 $AB^2 = \frac{90 \times 90}{106} = \frac{90 \times 90}{106} = \frac{8100}{106} = \frac{4050}{53}$

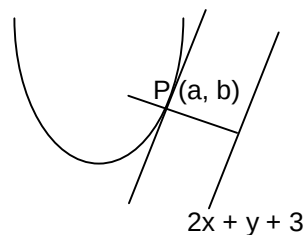


65. Let $P(a, b)$ be a point on the curve $y = 2x^2 + 6x + 5$ nearest to the line $2x + y + 3 = 0$, then the equation of the normal to the curve at P is

- (A) $x + 2y - 4 = 0$
 (B) $2x + y + 4 = 0$
 (C) $2x - y + 3 = 0$
 (D) $x - 2y + 4 = 0$

Ans. D

Sol. Let $y = 2x^2 + 6x + 5$
 $y' = 4x + 6$
 $4a + 6 = -2$
 $4a = -8$
 $a = -2$
 $y = 2(-2)^2 + 6(-2) + 5 = 8 - 12 + 5 = 1$
 $P(-2, 1)$ equation of normal at P
 $y - 1 = \frac{1}{2}(x + 2)$; $2y - 2 = x + 2$; $x - 2y + 4 = 0$



66. In any $\triangle ABC$ which of the following inequalities holds true?

- (A) $\sin \frac{A}{2} \cdot \sin \frac{B}{2} \cdot \sin \frac{C}{2} \leq \frac{1}{8}$
 (B) $\cos \frac{A}{2} \cdot \cos \frac{B}{2} \cdot \cos \frac{C}{2} \leq \frac{3\sqrt{3}}{8}$
 (C) $\cos^2 \frac{A}{2} + \cos^2 \frac{B}{2} + \cos^2 \frac{C}{2} \leq \frac{9}{4}$
 (D) All of the above

Ans. D

Sol. Obvious solutions

67. If the equation $\cos^4 \theta + \sin^2 \theta + \lambda = 0$ has real solution for θ , then λ lies in the interval

- (A) $\left[-1, -\frac{3}{4}\right]$
 (B) $\left[0, \frac{1}{2}\right]$
 (C) $(-1, 0)$
 (D) not possible

Ans. A

Sol. $\cos^2 \theta (1 - \sin^2 \theta) + \sin^2 \theta + \lambda = 0$

$$1 - \sin^2 \theta \cos^2 \theta + \lambda = 0$$

$$1 - \frac{1}{4}(\sin 2\theta)^2 + \lambda = 0 ; \lambda = \frac{1}{4}(\sin 2\theta)^2 - 1$$

$$\therefore 0 \leq \frac{1}{4} \sin^2 2\theta \leq \frac{1}{4} ; -1 \leq \frac{1}{4} \sin^2 2\theta - 1 \leq -\frac{3}{4}$$

$$\Rightarrow \lambda \in \left[-1, -\frac{3}{4} \right]$$

68. For some $\theta \in \left(0, \frac{\pi}{2} \right)$ if eccentricity of the hyperbola $x^2 - y^2 \operatorname{cosec}^2 \theta = 8$ is $\sqrt{7}$ times the eccentricity of the ellipse $x^2 + y^2 \operatorname{cosec}^2 \theta = 4$, then the length of the latus rectum of the ellipse is

(A) $\frac{8}{3}$

(B) $\frac{3}{2}$

(C) $\frac{8}{\sqrt{3}}$

(D) $8\sqrt{3}$

Ans. B

Sol. Hyperbola $\frac{x^2}{8} - \frac{y^2}{8 \sin^2 \theta} = 1 ; e_n^2 = 1 + \frac{8 \sin^2 \theta}{8} ; e_n = \sqrt{1 + \sin^2 \theta}$

Ellipse $\frac{x^2}{4} + \frac{y^2}{4 \sin^2 \theta} = 1 ; e_e^2 = 1 - \frac{4 \sin^2 \theta}{4} = 1 - \sin^2 \theta ; e_e = \sqrt{1 - \sin^2 \theta}$

$$e_n = \sqrt{7} e_e \Rightarrow 1 + \sin^2 \theta = 7(1 - \sin^2 \theta) = 8 \sin^2 \theta = 6$$

$$\Rightarrow \sin^2 \theta = \frac{3}{4} \text{ latus rectum of ellipse} = \frac{2b^2}{a} = \frac{3}{2}$$

69. Rectangle ABCD has AB = 5 and BC = 4, point E lies on AB so that EB = 1 point G lies on BC so that CG = 1 and point F lies on CD so that DF = 2, segment AG and AC intersect EF at Q and P respectively that ratio $\frac{PQ}{EF}$ is

(A) $\frac{\sqrt{13}}{16}$

(B) $\frac{\sqrt{2}}{13}$

(C) $\frac{9}{82}$

(D) $\frac{10}{91}$

Ans. D

Sol. Equation of AG

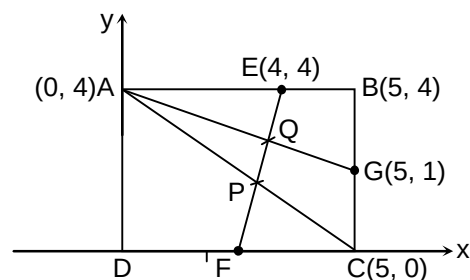
$$y - 4 = \frac{4-1}{0-5}(x-0) ; y - 4 = -\frac{3}{5}(x)$$

$$5y - 20 = -3x ; y = -\frac{3}{5}x + 4$$

Equation AC

$$y = -\frac{4}{5}x + 4$$

$$\text{Equation of EF, } y = 2x - 7 \Rightarrow \frac{PQ}{EF} = \frac{10}{91}$$



70. Greatest and least value of $(\sin^{-1}x)^2 + (\cos^{-1}x)^2$ are respectively

(A) $\frac{\pi^2}{4}$ and $-\frac{\pi^2}{4}$

(B) $\frac{\pi}{2}$ and $-\frac{\pi}{4}$

(C) $\frac{\pi^2}{4}$ and 0

(D) $\frac{5\pi^2}{4}$ and $\frac{\pi^2}{8}$

Ans. D

Sol. $(\sin^{-1}x)^2 + (\cos^{-1}x)^2 = (\sin^{-1}x + \cos^{-1}x)^2 - 2\sin^{-1}x\cos^{-1}x = \frac{\pi^2}{4} - 2\sin^{-1}x\left(\frac{\pi}{2} - \sin^{-1}x\right)$

$$\Rightarrow \text{Least value } \frac{\pi}{8}, \text{ greatest value } \frac{5\pi^2}{4}$$

71. The centre of the circle passing through (0, 2) and touching the parabola $y = 2x^2$ at (2, 2) is

(A) $\left(2, \frac{5}{2}\right)$

(B) $\left(-2, \frac{5}{2}\right)$

(C) (12, 4)

(D) (4, 12)

Ans. B

Sol. Equation of tangent at (4, 2) to $y = 2x^2$ is $\frac{y+2}{2} = 2(2 \cdot x)$

$$\Rightarrow y + 2 = 8x ; 8x - y - 2 = 0$$

Equation family of circle touching the parabola at (2, 2) is $(x - 2)^2 + (y - 2)^2 + \lambda(8x - y - 2) = 0$

If this passes through (0, 2) ; $4 + 0 + \lambda(-2, -2) = 0 ; 4 + \lambda(-4) = 0, \lambda = 1$

$$\Rightarrow (x - 2)^2 + (y - 2)^2 + 8x - y - 2 = 0 ; x^2 + y^2 - 4x - 4y + 8 + 8x - y - 2 = 0$$

$$x^2 + y^2 + 4x - 5y + 6 = 0 \text{ centre } \left(-2, \frac{5}{2}\right)$$

72. Let L_1 be a tangent to the parabola $y^2 = 4(x + 2)$ and L_2 be a tangent to the parabola $y^2 = 8(x + 3)$ such that L_1 and L_2 intersect at right angles, then L_1 and L_2 meet on the straight line

(A) $x + 4 = 0$

(B) $x - 4 = 0$

(C) $y + 4 = 0$

(D) $y - 4 = 0$

Ans. A

Sol. Equation of tangent to $y^2 = 4(x + 2)$ is

$$y = m(x + 2) + \frac{1}{m} \quad \dots (1)$$

$$\text{Equation of tangent to } y^2 = 8(x + 3) \text{ is } y = -\frac{1}{m}(x + 3) + \frac{2}{-\frac{1}{m}}$$

$$y = -\frac{1}{m}(x + 3) - 2m \quad \dots (2)$$

$$\text{Solving for point of intersection } m(x + 2) + \frac{1}{m} = -\frac{1}{m}(x + 3) - 2m$$

$$m(x + 4) + \frac{1}{m} \cdot (x + 4) = 0 ; \left(m + \frac{1}{m}\right)(x + 4) = 0 \Rightarrow x + 4 = 0$$

73. A ray of light coming from the point $(3, 3\sqrt{3})$ is incident at an angle 30° on the line $x = 2$ at the point A. The ray gets reflected on the line $x = 2$ and meets x-axis at the point B. The line AB passes through the point

(A) $(1, \sqrt{3})$

(B) $(2, 2\sqrt{3})$

(C) $(-2, 2\sqrt{3})$

(D) $(2, -2\sqrt{3})$

Ans. B

Sol. Equation of AP ; $y - 3\sqrt{3} = \sqrt{3}(x - 3)$

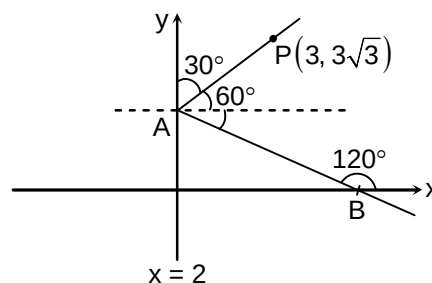
$$y = \sqrt{3}x$$

$$A(2, 2\sqrt{3})$$

Equation of AB ; $y - 2\sqrt{3} = -\sqrt{3}(x - 2)$

$$y - 2\sqrt{3} = -\sqrt{3}x + 2\sqrt{3}$$

$$\sqrt{3}x + y - 4\sqrt{3} = 0$$



74. Which of the following points lies on the locus of the foot of perpendicular drawn upon any tangent to the ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 1$ from any of its foci

(A) $(1, 2)$

(B) $(\sqrt{2}, 2)$

(C) $(1, 2\sqrt{2})$

(D) $(3, 3)$

Ans. C

Sol. So, locus is its auxiliary circle $x^2 + y^2 = 9$

75. If tangent to the hyperbola $\frac{x^2}{4} - y^2 = 1$ cut the circle $x^2 + y^2 = 16$ at two distinct point A and B, then the locus of mid-point of A and B is

(A) $x^2 - y^2 = \sqrt{4x^2 - y^2}$

(B) $x^2 + y^2 = \sqrt{4x^2 + y^2}$

(C) $x^2 y^2 = \sqrt{x^2 + y^2}$

(D) $x^2 + y^2 = \sqrt{4x^2 - y^2}$

Ans. D

Sol. The equation of tangent to hyperbola

$$y = mx \pm \sqrt{4m^2 - 1} \quad \dots (1)$$

Let mid-point of AB is (h, k) so, equation chord AB is $T = S_1$

$$\Rightarrow hx + ky = h^2 + k^2$$

$$y = -\frac{h}{k}x + \frac{(h^2 + k^2)}{k} \quad \dots (2)$$

By equation (1) and (2), we get $m = -\frac{h}{k}$

$$\frac{h^2 + k^2}{k} = \sqrt{4m^2 - 1} ; \frac{h^2 + k^2}{k} = \sqrt{4 \cdot \frac{h^2}{k^2} - 1}$$

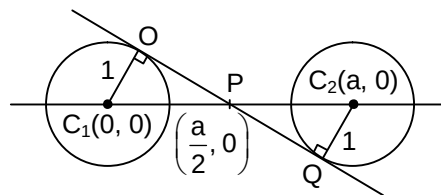
$$\Rightarrow h^2 + k^2 = \sqrt{4h^2 - k^2} \Rightarrow x^2 + y^2 = \sqrt{4x^2 - y^2}$$

76. The length of transverse common tangent of circles $x^2 + y^2 = 1$ and $(x - a)^2 + y^2 = 1$ is $2\sqrt{3}$, then the value of a is (where $a > 2$)

- (A) 3
(B) 4
(C) $\sqrt{5}$
(D) $\sqrt{6}$

Ans. B

Sol. $OQ = \sqrt{a^2 - (2)^2} = \sqrt{a^2 - 4} = 2\sqrt{3}$
 $a^2 - 4 = 12$
 $a^2 = 16 \Rightarrow a = 4$



77. The number of solution of $2\sin x = 3^{x+1} + 3^{1-x}$ is

- (A) one solution
(B) infinite solution
(C) zero solution
(D) cannot be calculated

Ans. C

Sol. $2\sin x = 3\left(3^x + \frac{1}{3^x}\right) = \text{R.H.S} \geq 6$
 $\text{L.H.S} \leq 2$, no solution

78. The range of $\frac{1}{4|\sin x| - 3}$ will be

- (A) $\left(-\infty, \frac{1}{3}\right] \cup [1, \infty)$
(B) $\left(-\infty, -\frac{1}{3}\right] \cup [1, \infty)$
(C) $[2, \infty)$
(D) $(-\infty, \infty)$

Ans. B

Sol. $0 \leq |\sin x| \leq 1$
 $0 \leq 4|\sin x| < 2$
 $-3 \leq 4|\sin x| - 3 < 1$
 $\Rightarrow \frac{1}{4|\sin x| - 3} \in \left(-\infty, -\frac{1}{3}\right] \cup [1, \infty)$

79. If $\sec \theta - \tan \theta = a$ and $\operatorname{cosec} \theta + \cot \theta = b$, then which relation is true?

- (A) $b - a = 1 + ab$
 (B) $ab = a - b$
 (C) $1 - ab = a - b$
 (D) all of the above

Ans. A

Sol. $ab + 1 = (\sec \theta - \tan \theta)(\operatorname{cosec} \theta + \cot \theta) + 1 = (\tan \theta + \cot \theta) - (\sec \theta - \operatorname{cosec} \theta)$
 $\Rightarrow (\operatorname{cosec} \theta + \cot \theta) - (\sec \theta - \tan \theta) = b - a$

80. If $(x - 1)(x^2 + 2) > 0$, then the value of $\sin\left(\frac{1}{2}\tan^{-1}\left(\frac{2x}{1-x^2}\right) - \tan^{-1}x\right)$ is

- (A) 1
 (B) -1
 (C) 0
 (D) 2

Ans. B

Sol. $(x - 1)(x^2 + 2) > 0 \Rightarrow x > 1$
 $\therefore \sin\left(\frac{1}{2}\tan^{-1}\frac{2x}{1-x^2} - \tan^{-1}x\right) = \sin\left(\frac{1}{2}(-\pi + 2\tan^{-1}x) - \tan^{-1}x\right)$
 $= \sin\left(-\frac{\pi}{2}\right) = -1$

SECTION - B (Numerical Answer Type)

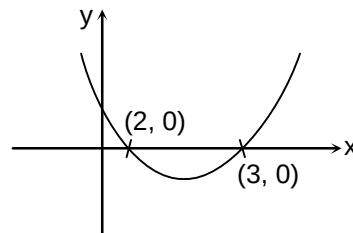
This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE**. For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXX.XX).

81. If the line $2x - y = \ell$ and $x + 2y = m$ touch the curve $y = x^2 - 5x + 6$ at the points where the curve intersect the x-axis, then $\frac{\ell}{m}$ is equal to

Ans. 00003.00

Sol. $y = x^2 - 5x + 6$ cut x-axis at (2, 0) and (3, 0)

$$\Rightarrow \ell = 6, m = 2 \text{ so, } \frac{\ell}{m} = 3$$



82. If $A + B + C = \frac{\pi}{2}$, then $\frac{1}{\cot A \cot B} + \frac{1}{\cot B \cot C} + \frac{1}{\cot C \cot A}$ is equal to

Ans. 00001.00

Sol. If $A + B + C = \frac{\pi}{2}$, $\cot A \cot B \cot C = \cot A + \cot B + \cot C$

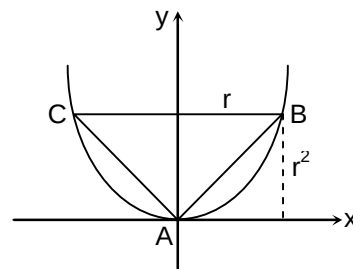
83. All three vertices of $\triangle ABC$ lie on the parabola $y = x^2$ with A at origin and BC parallel to the x-axis. The area of the triangle is 64. What is the length of BC

Ans. 00008.00

Sol. Area = $\frac{(2r) \cdot r^2}{2} = r^3 = 64$

$$r = 4$$

$$BC = 8$$



84. A line parallel to the straight line $2x - y + 1 = 0$ is tangent to the hyperbola $\frac{x^2}{4} - y^2 = 1$ at the point $p(h, k)$, then $h^2 + 11k^2$ is

Ans. 00005.00

Sol. Equation of tangent at $p(h, k)$ to the hyperbola is $\frac{x \cdot h}{4} - yk = 1$, slope of the tangent = $\frac{h}{4k} = 2$

$\Rightarrow h = 8k$ and (h, k) lies on the curve

$$\frac{h^2}{4} - k^2 = 1 ; \frac{8^2 k^2}{4} - k^2 = 1 ; 15k^2 = 1$$

$$\Rightarrow k^2 = \frac{1}{15} \text{ so, } h^2 = 64 \cdot k^2 = \frac{64}{15}$$

$$h^2 + 11k^2 = \frac{64}{15} + \frac{11}{15} = \frac{75}{15} = 5$$

85. If the line $y = mx + 2$ is tangent to both the parabola $y^2 = 8x$ and $x^2 = 4by$, then $|b|$ is equal to

Ans. 00002.00

Sol. Equation of tangent to $y^2 = 8x$ is $y = mx + \frac{a}{m}$
 $\Rightarrow y = mx + \frac{2}{m}$ (1)

So, $y = mx + 2$ is given on comparing $\frac{2}{m} = 2$, $m = 1$

So, line $y = x + 2$ is tangent to $x^2 = 4by$
 $\Rightarrow x^2 - 4b(x + 2) = 0$; $x^2 - 4bx - 8b = 0$
 $\Rightarrow D = 0$, $16b^2 + 32b = 0$; $b \neq 0$, $b = -2$

86. If $2x + 3y = 15$ is a tangent to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{9} = 1$ for some $a \in \mathbb{R}$, then square of the distance between the foci of the ellipse is

Ans. 00108.00

Sol. $2x + 3y = 15$; $3y = -2x + 15$
 $y = \frac{-2}{3}x + 5$ is tangent to ellipse $\frac{x^2}{a^2} + \frac{y^2}{9} = 1$
 So, $c^2 = a^2m^2 + b^2$; $25 = a^2 \frac{4}{9} + 9$
 $a^2 = \frac{16 \times 9}{4} \Rightarrow a = \frac{4 \times 3}{2} = 6$
 $b^2 = a^2(1 - e^2) \Rightarrow 9 = 36(1 - e^2)$; $\frac{9}{36} = \frac{1}{4} = 1 - e^2$
 $e^2 = 1 - \frac{1}{4} = \frac{3}{4}$; $e = \frac{\sqrt{3}}{2}$
 Distance between focus = $2ae = 108$

87. Number of root of the equation is $|\sin x \cos x| + \sqrt{\tan^2 x + \cot^2 x + 2} = \sqrt{2}$ where $x \in [0, 4\pi]$

Ans. 00000.00

Sol. $|\sin x \cos x| + \frac{1}{|\sin x \cos x|} \geq 2$, so it has no solution

88. The number of value of x which satisfy $\cos^{-1} \frac{1}{x} + \cos^{-1} x + \sin^{-1} \frac{1}{x} + \sin^{-1} x$

Ans. 00002.00

Sol. Since, $x, \frac{1}{x} \in [-1, 1] \Rightarrow x = \pm 1$ only

89. If $\tan 40^\circ = a$, then $\frac{\tan 140^\circ - \tan 130^\circ}{1 + \tan 140^\circ \tan 130^\circ}$ is $\frac{1 - ka^2}{2a}$, then k is equal to

Ans. 00001.00

Sol.
$$\frac{\tan(180^\circ - 40^\circ) - \tan(90^\circ + 40^\circ)}{1 + \tan(180^\circ - 40^\circ)\tan(90^\circ + 40^\circ)} = \frac{-\tan 40^\circ + \cot 40^\circ}{1 + \tan 40^\circ \cot 40^\circ} = \frac{-a + \frac{1}{a}}{1 + a \cdot \frac{1}{a}} = \frac{1 - a^2}{2a}$$

90. If the line $y = ax + b$ is a common tangent to the hyperbola $\frac{x^2}{144} - \frac{y^2}{100} = 1$ and circle $x^2 + y^2 = 64$, then $\frac{80a^2}{41}$ is

Ans. 00004.00

Sol.
$$\begin{aligned} b^2 &= 144a^2 - 100 ; b^2 = 64(1 + a^2) \\ \Rightarrow 144a^2 - 100 &= 64 + 64a^2 \\ 80a^2 &= 164 \\ a^2 &= \frac{164}{80} = \frac{41}{20} \end{aligned}$$